ME 160
WORKING DRAWING

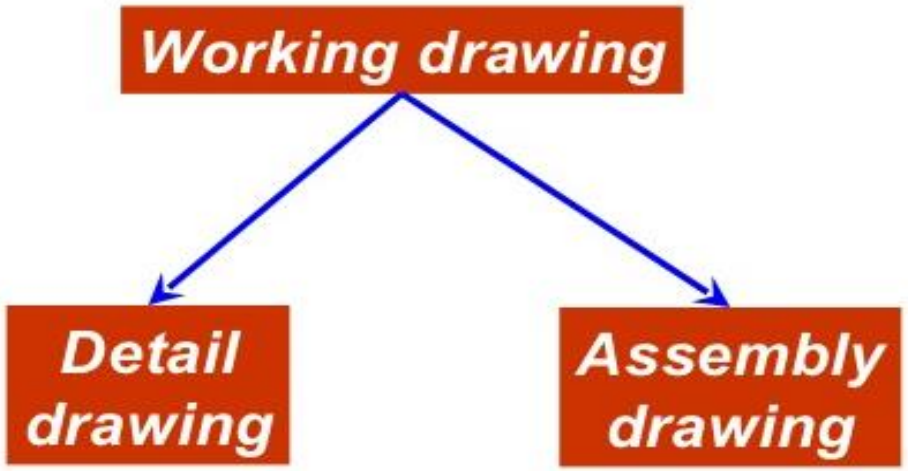
Working Drawing





DEFINITION

A working drawing is a set of drawings used during the process of making a product



DEFINITION

- ***Detail drawing*** is a ***multiview representation*** of a single part with ***dimensions and notes***.

- ***Assembly drawing*** is a drawing of various parts of a machine or structure assembled in their relative working positions.

DETAIL DRAWING

INFORMATION IN DETAIL DRAWING

1. General information → Title block

2. Part' s information

2.1 Shape description → Object's views

2.2 Size description → Object's views

2.3 Specifications → Notes

DETAIL DRAWING

GENERAL INFORMATION

- Name of company
- Title of drawing (usually part's name)
- Drawing sheet number
- Name of drafter, checker
- Relevant dates of action
  (drawn, checked, approved etc.)
- Revision table
- Unit
- Scale
- Method of projection

DETAIL DRAWING

PART' S INFORMATION

Shape	❖ **Orthographic drawing** ❖ Pictorial drawing
Size	❖ **Dimensions** and Tolerances
Specifications	❖ **Part number, name, number required** ❖ **Type of material used** ❖ **General notes** ❖ Heat treatment ❖ Surface finish ❖ General tolerances

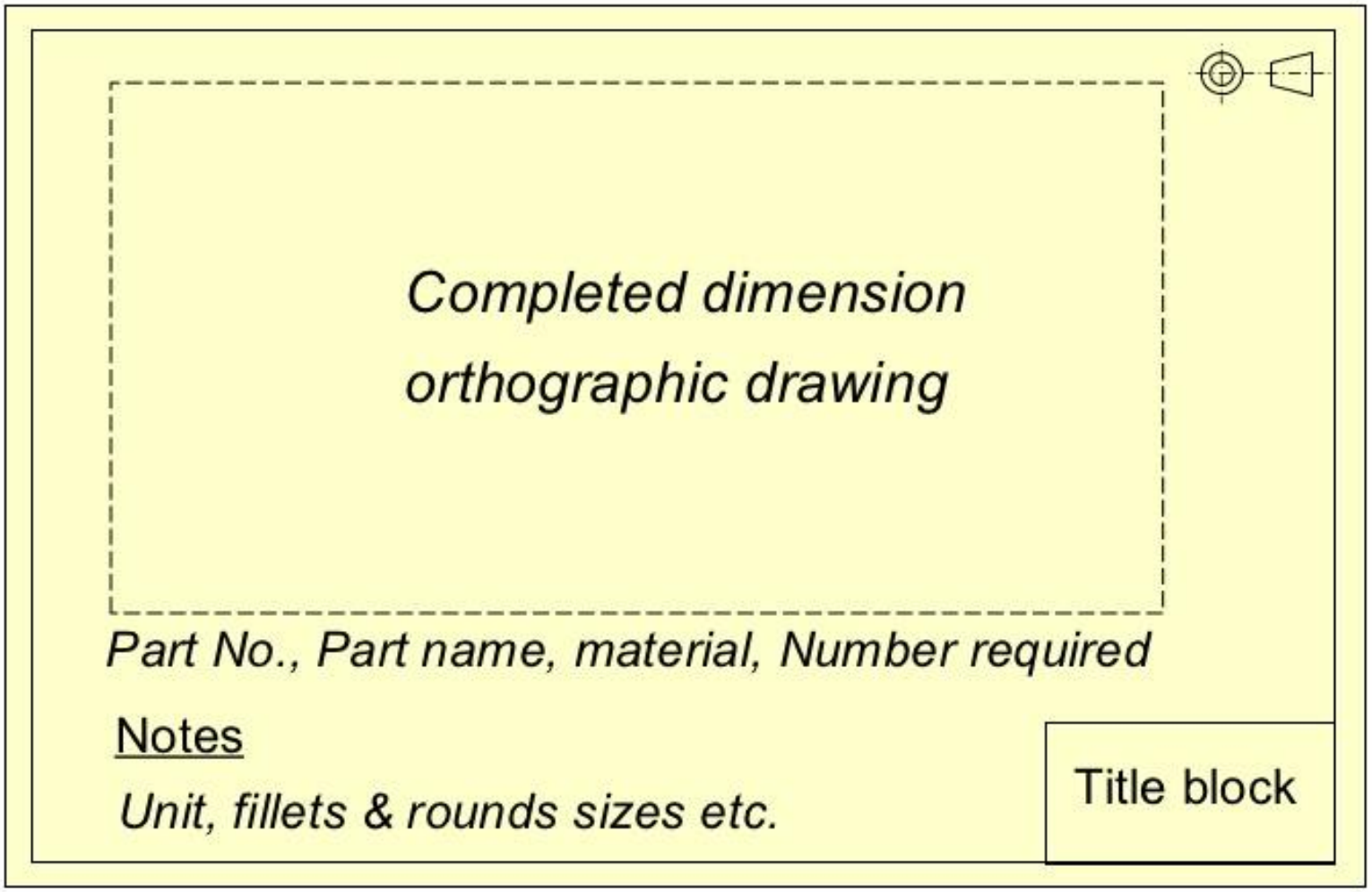
Detail Drawing

RECOMMENDED PRACTICE

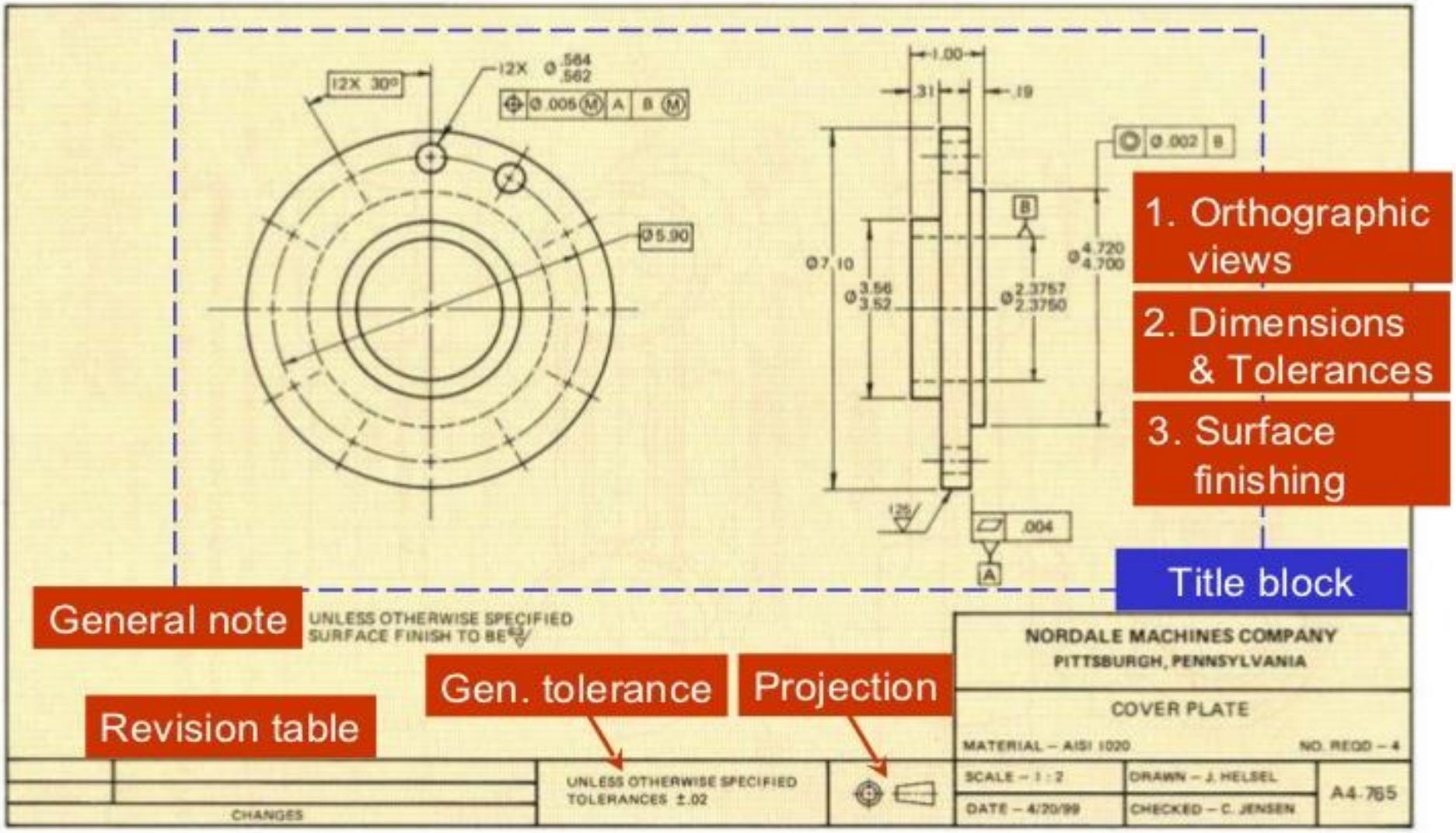
- Draw **one part** to **one sheet** of paper.
- If not the case,
  - apply ***enough spacing*** between parts.
  - draw all parts using the ***same scale***.
    Otherwise, the scale should be clearly note under each part's drawing.
- Standard parts such as ***bolt***, ***nut***, ***pin***, ***bearing*** do not require detail drawings.

PLACING AN INFORMATION

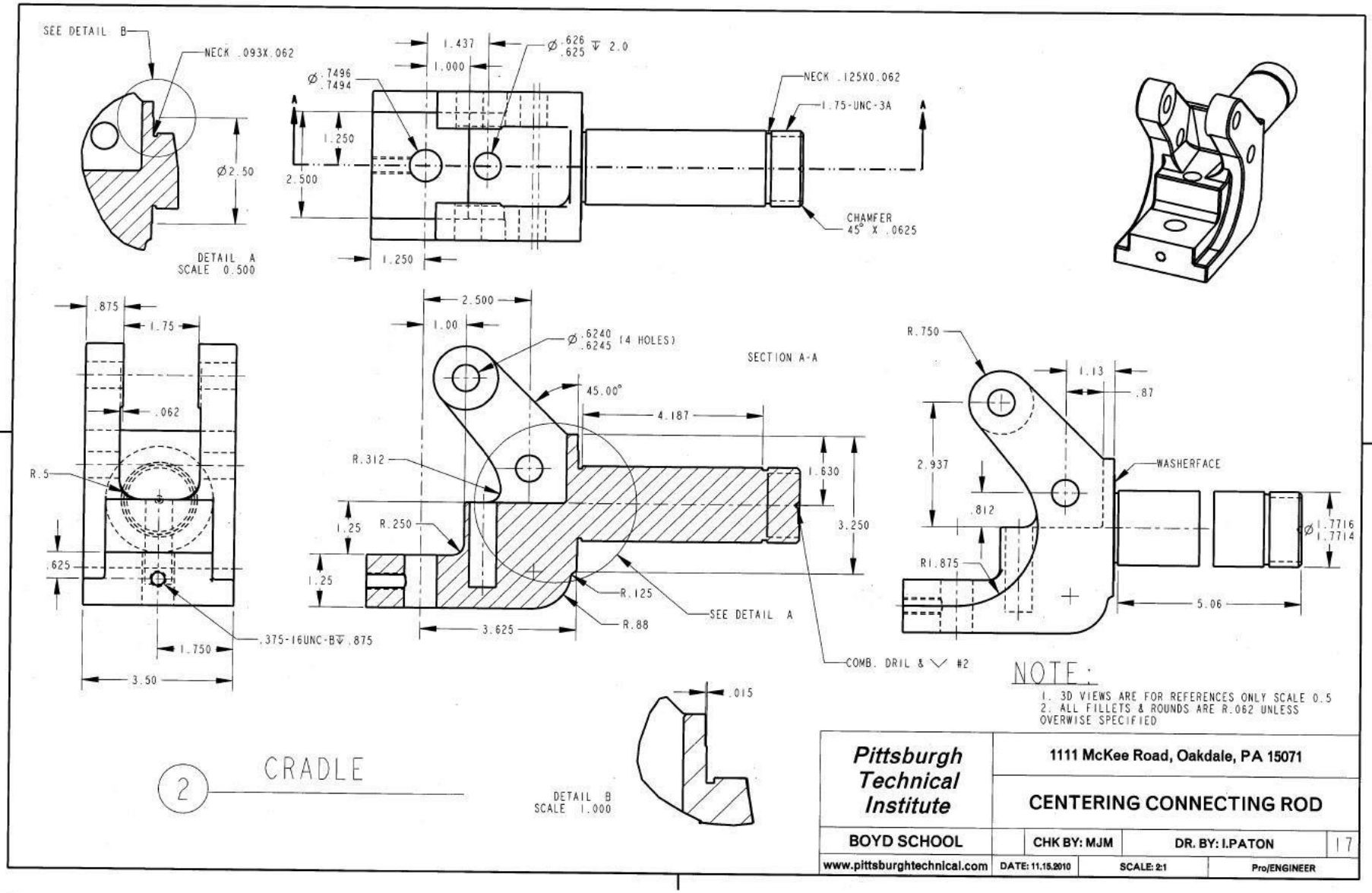
(This course)



EXAMPLE : Interpreting detail drawing



Detail Drawing



Assembly Drawing



TYPES OF ASSEMBLY DRAWING

1. Exploded assembly drawings

The parts are separately display, but they are aligned according to their assembly positions and sequences.

2. General assembly drawings.

All parts are drawn in their working position.

3. Detail assembly drawings

All parts are drawn in their working position with a completed dimensions.

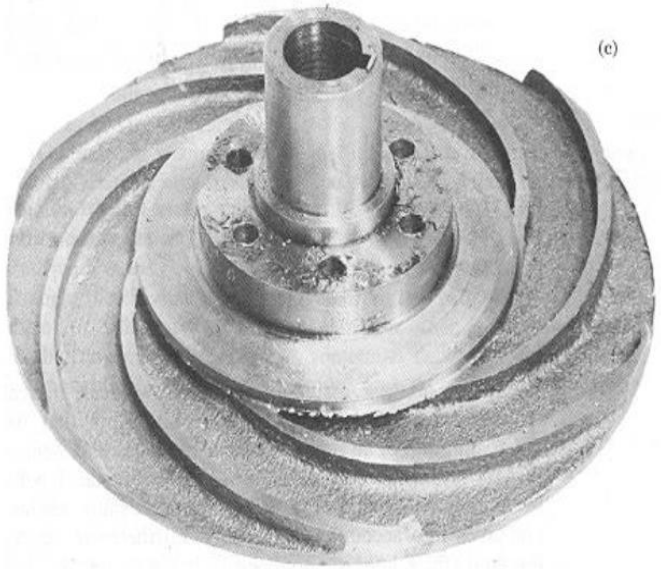
Exploded Assembly



Centrifugal pump impeller (shrouded)

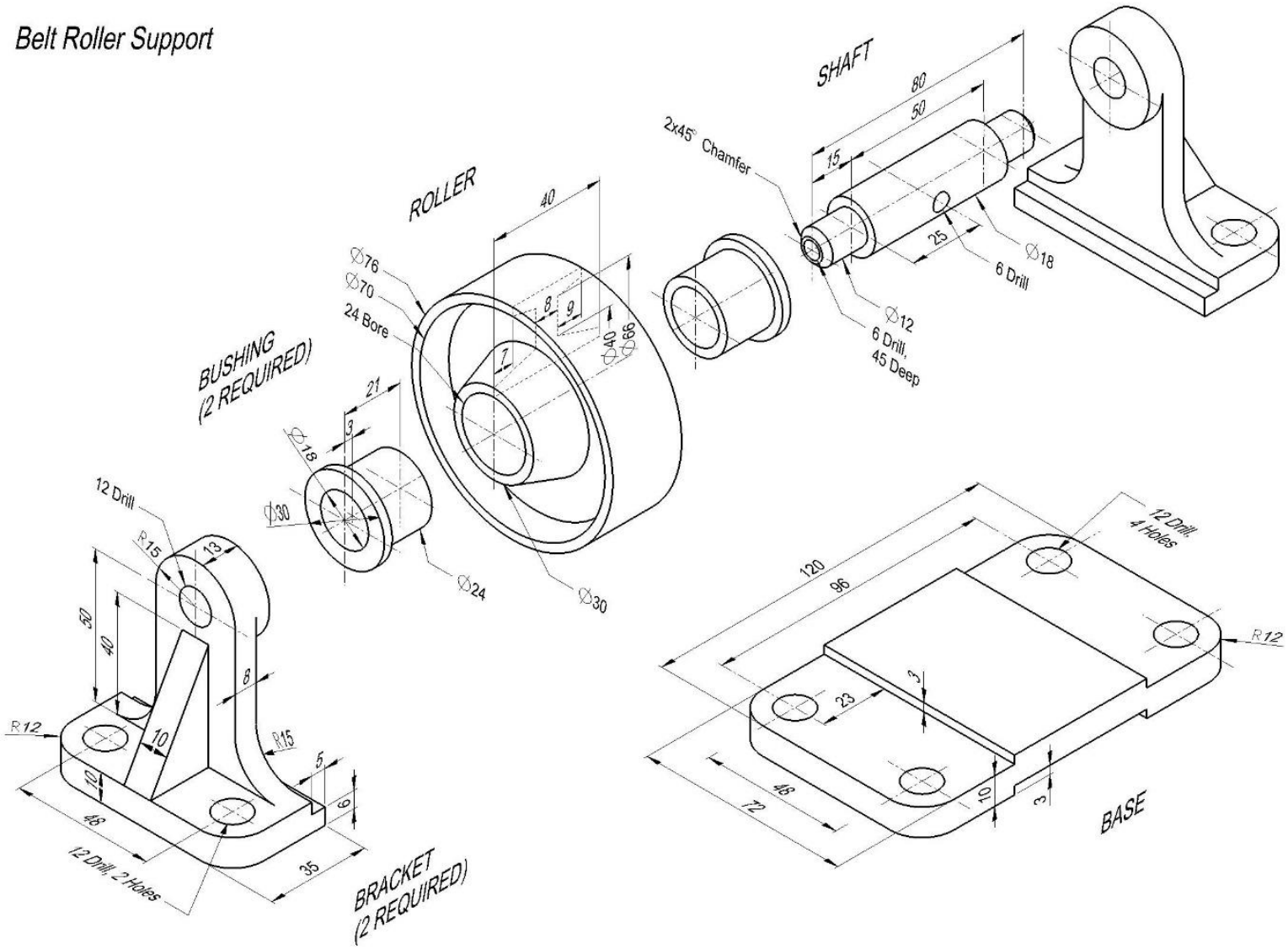


Centrifugal pump impeller (unshrouded)



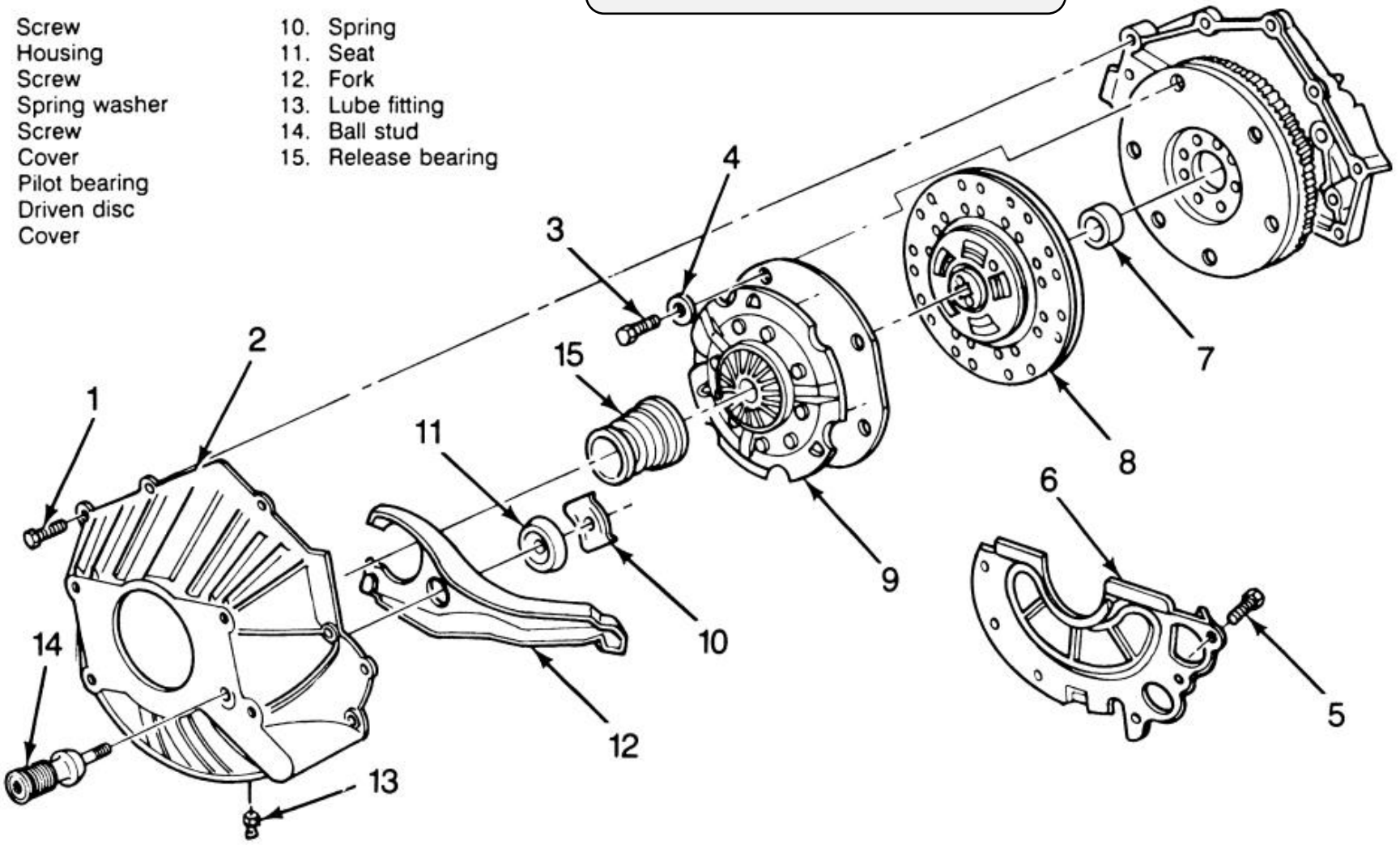


Exploded Assembly

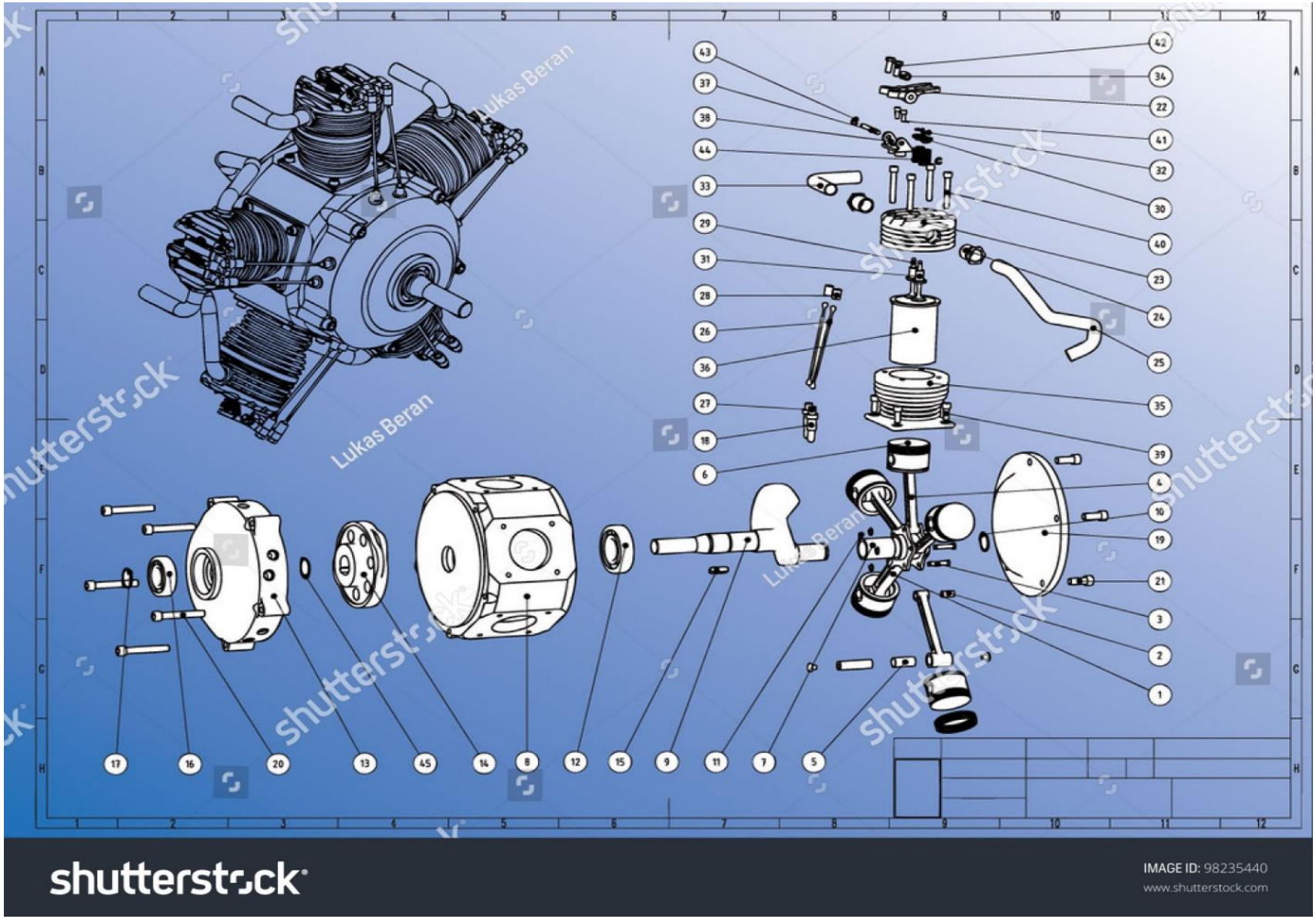


Exploded Assembly

1. Screw
2. Housing
3. Screw
4. Spring washer
5. Screw
6. Cover
7. Pilot bearing
8. Driven disc
9. Cover
10. Spring
11. Seat
12. Fork
13. Lube fitting
14. Ball stud
15. Release bearing

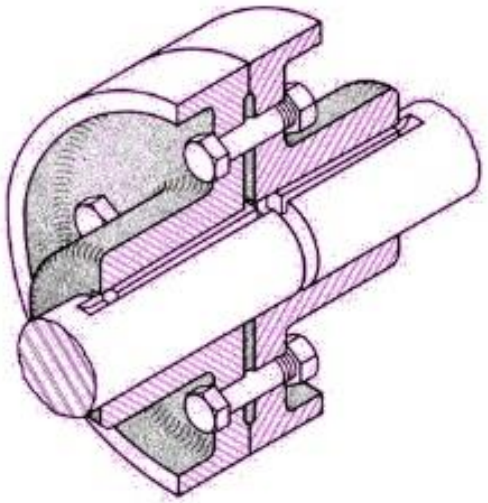


Exploded view

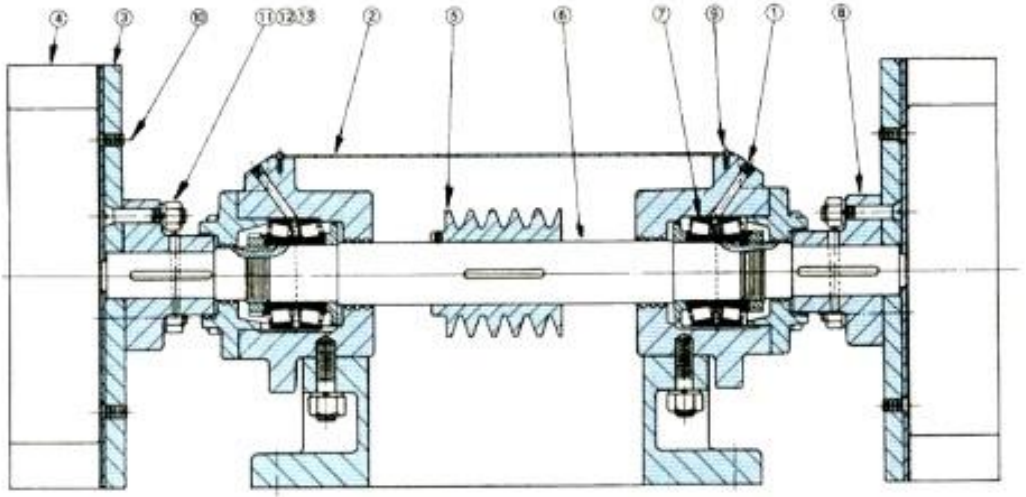


2. GENERAL ASSEMBLY

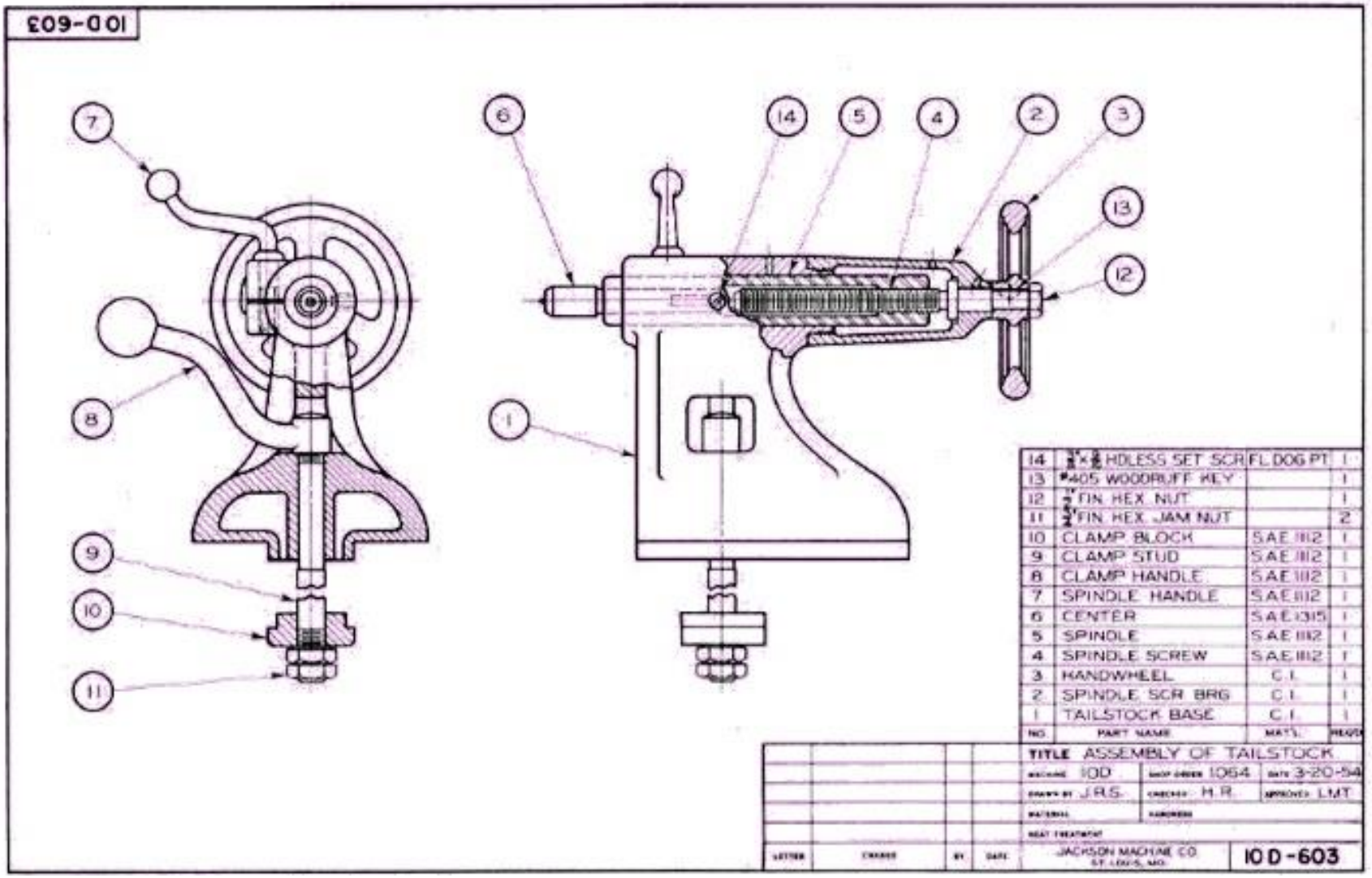
Pictorial



Orthographic



2. GENERAL ASSEMBLY



NO	PART NAME	MAT'L	REQ'D
14	3/8"x3/8" HDLESS SET SCR FL DOG PT		1
13	#405 WOODRUFF KEY		1
12	3/4" FIN HEX NUT		1
11	3/4" FIN HEX JAM NUT		2
10	CLAMP BLOCK	SAE 1112	1
9	CLAMP STUD	SAE 1112	1
8	CLAMP HANDLE	SAE 1112	1
7	SPINDLE HANDLE	SAE 1112	1
6	CENTER	SAE 1315	1
5	SPINDLE	SAE 1112	1
4	SPINDLE SCREW	SAE 1112	1
3	HANDWHEEL	C.I.	1
2	SPINDLE SCR BRG	C.I.	1
1	TAILSTOCK BASE	C.I.	1

TITLE ASSEMBLY OF TAILSTOCK

MACHINE 10D | SHOP ORDER 1064 | DATE 3-20-54

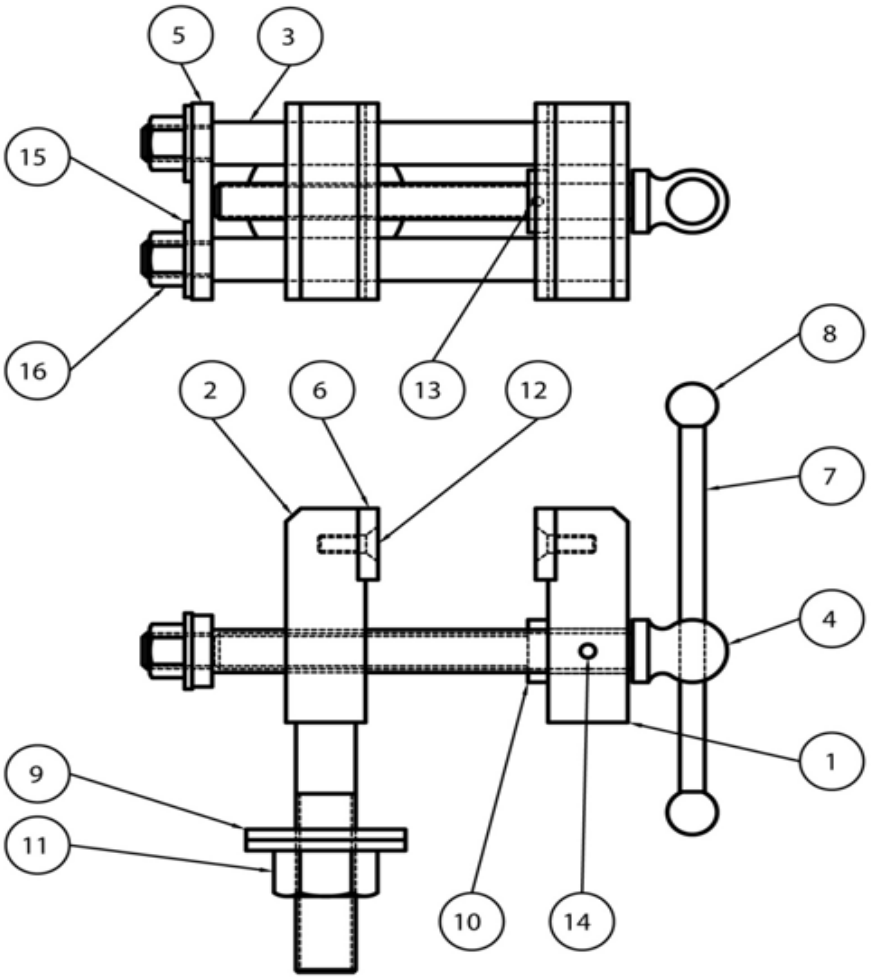
DRAWN BY J.R.S. | CHECKED H.R. | APPROVED L.M.T.

MATERIAL | HARDNESS

HEAT TREATMENT

LETTER | CHANGE | BY | DATE | JACKSON MACHINE CO. ST. LOUIS, MO. | 10D-603

GENERAL ASSEMBLY

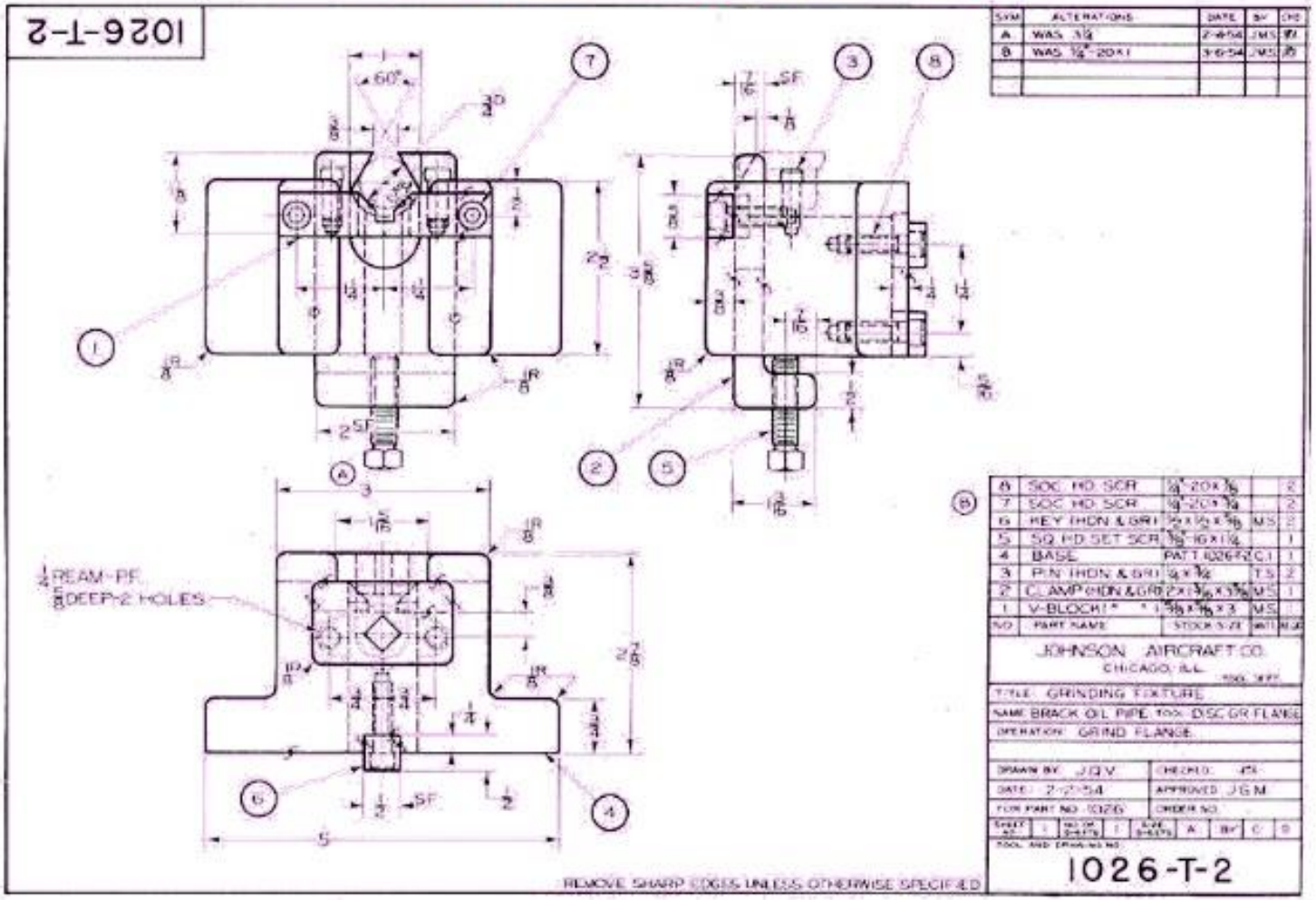


16	GUIDE NUT	2	-
15	GUIDE WASHER	2	-
14	JAW PIN	2	-
13	COLLAR PIN	1	-
12	FLAT HEAD SCREW	4	-
11	JAW NUT	1	-
10	COLLAR	1	1020 ST
9	JAW WASHER	2	-
8	REMOVABLE BALL	2	CRS
7	HANDLE	1	CRS
6	JAW INSERT	2	1020 ST
5	SPACER	1	CRS
4	SCREW	1	CRS
3	GUIDE BAR	2	1020 ST
2	STATIONARY JAW	1	CRS
1	MOVABLE JAW	1	CRS
PART #	PART NAME	REQ'D	MAT

DRAWN BY: KIRSTIE PLANTENBERG	SHEET #: 1	# OF SHEETS: 11	DATE:	SCALE: 1:2
PART NAME: MACHINIST'S VISE	PART #: -	MAT: -		REQ'D: 1

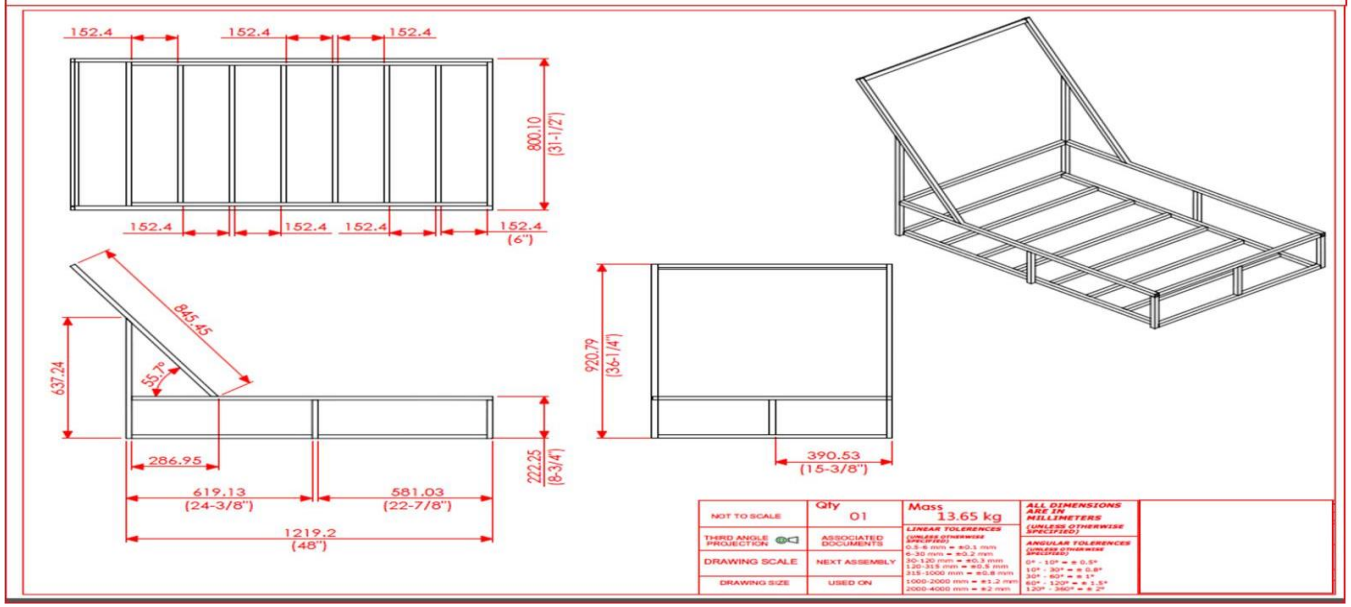
3. DETAILED ASSEMBLY

(working-drawing assembly)



DETAILED ASSEMBLY



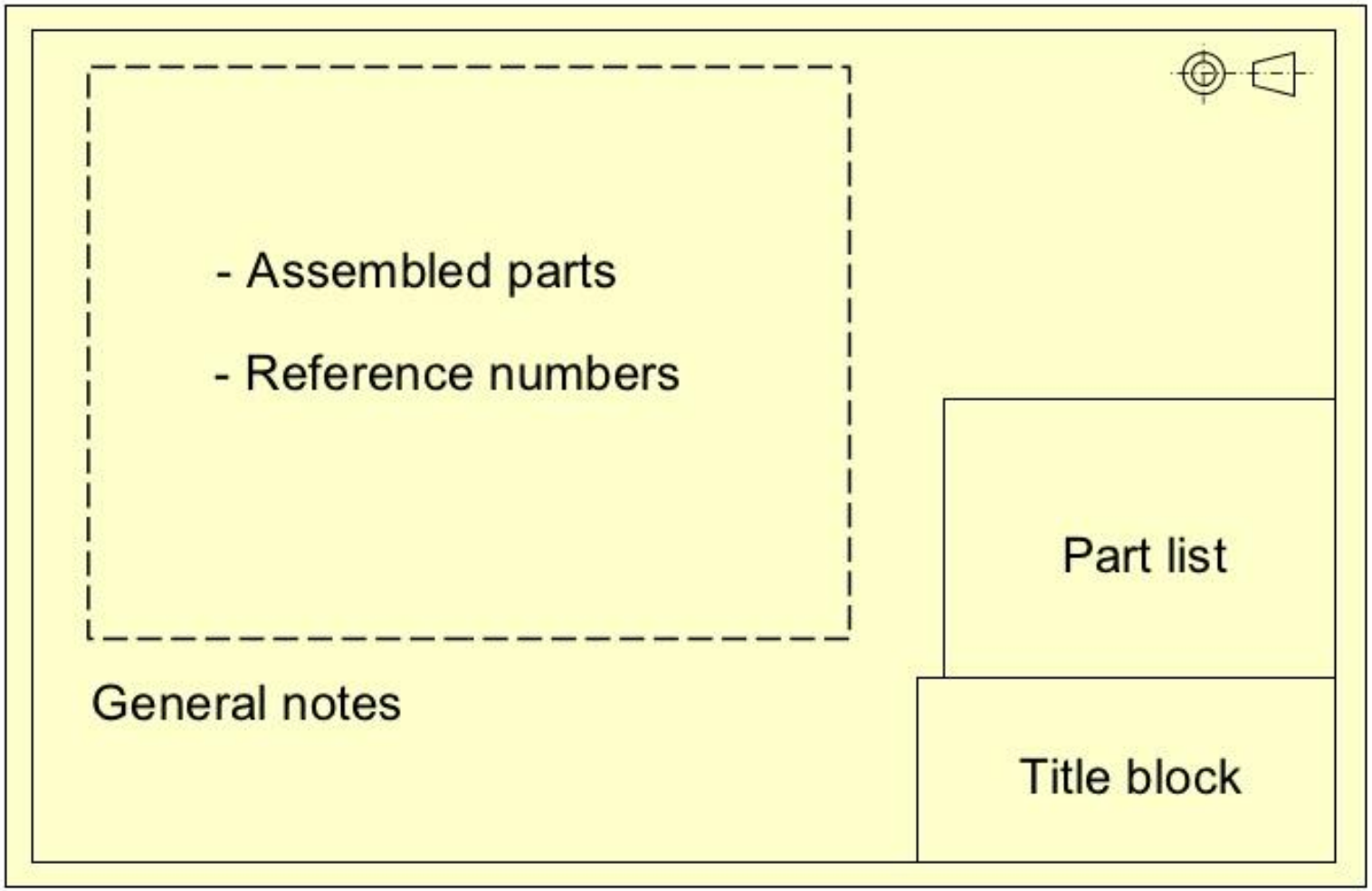


REQUIRED INFORMATION IN GENERAL ASSEMBLY DRAWING

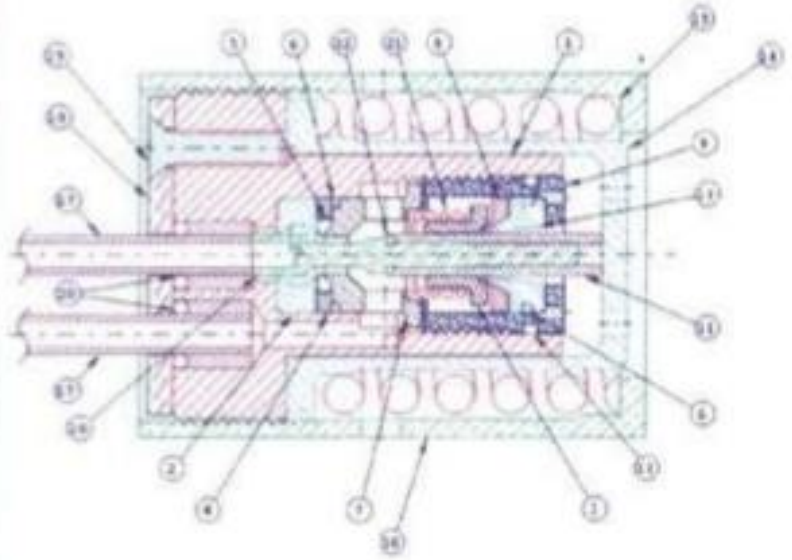
1. **All parts, drawn in their operating position.**
2. **Part list (or bill of materials, BOM)**
    1. Item number
    2. Descriptive name
    3. Material, MATL.
    4. Quantity required (per a unit of machine), QTY.
3. **Leader lines** with **balloons** around **part numbers.**
4. **Machining** and **assembly operations** and **critical dimensions** related to operation of the machine.

PLACING AN INFORMATION

(This course)



EXAMPLE : Another allowable place for BOM

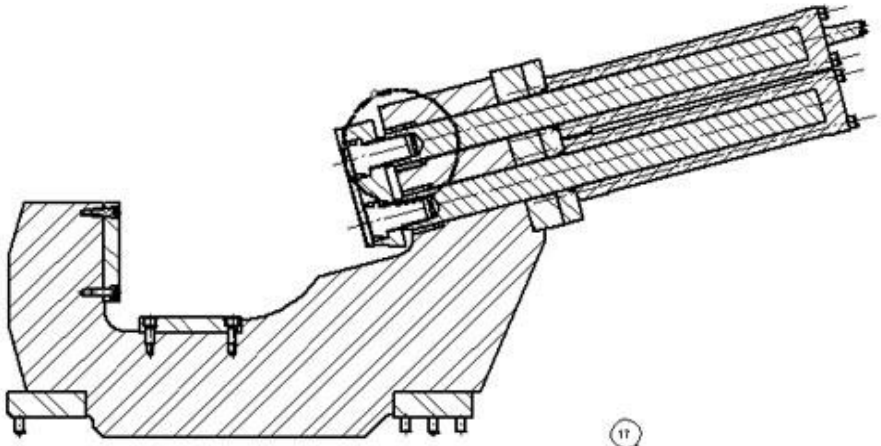


BILL OF MATERIAL

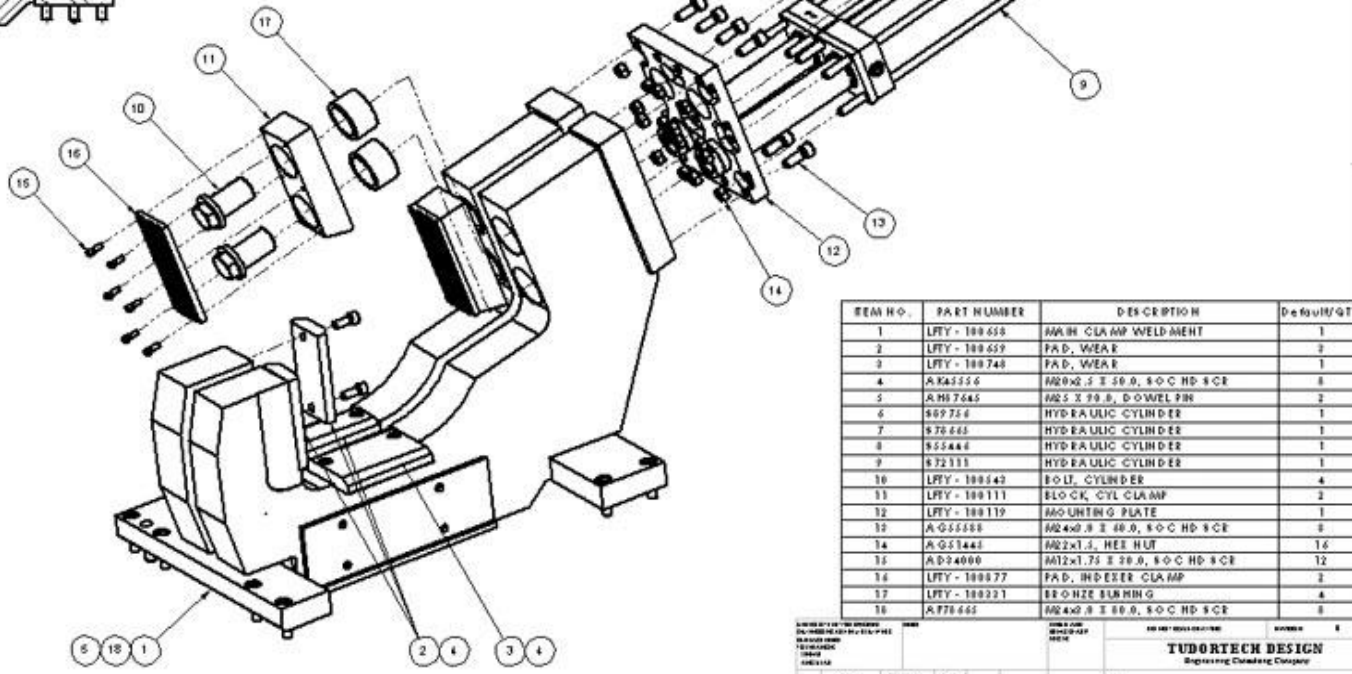
ITEM NO.	PART NO.	DESCRIPTION/TITLE	NO. REQD	REMARKS
1	BR2001	BODY	1	
2	BR2002	ARM SUPPORT	1	
3	BR2003	SEAL SUPPORT	1	
4	BR2004	SPACER RING	1	
5		O-RING .140 I.D. x .040 W	1	
6		O-RING .239 I.D. x .012 W	2	
7	BR2005	MANIFOLD	1	
8	BR2006	TEFLON SEAL	1	
9	BR2007	RETAINING NUT	1	
10	BR2008	POPPET STEM	1	
11	BR2009	POPPET SEALING SLEEVE	1	
12		O-RING .855 I.D. x .012 W	1	
13	BR2010	SEAL SUPPORT	1	
14	BR2011	CAGE 1	1	
15		SPRING	1	
16	BR2012	CAGE 2	1	
17	BR2013	PORT TUBE	2	
18	BR2014	CLAMPING DISC	1	
19		HEAD DOWN SCREW	1	
20		O-RING .987 I.D. x .939 W	2	
21		EXERCISER SPRING	1	
22		O-RING	1	

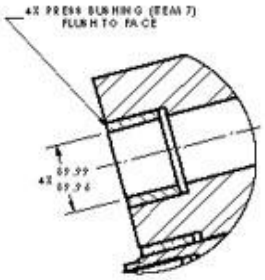
STONE ENGINEERING COMPANY
A DIVISION OF THE MAKO CORPORATION

DRAWN BY: TLH	DATE:	TOLERANCES:
DATE: 03/21/95		
SCALE: 3=1		
TITLE: HYDRAZINE REGULATOR ASSEMBLY		DRAWING NO. BR1001



SECTION A-A





DETAIL C
SCALE 1 : 4
SHOWN WITHOUT ITEMS 5 & 18
FOR CLARITY

ITEM NO.	PART NUMBER	DESCRIPTION	Default/QTY.
1	LFTY - 100658	MAIN CLAMP WELDMENT	1
2	LFTY - 100659	PAD, WEAR	2
3	LFTY - 100748	PAD, WEAR	1
4	AKA5556	M20x2.5 X 50.0, SOC HD SCR	8
5	AH67665	M25 X 90.0, DOWEL PIN	2
6	969756	HYDRAULIC CYLINDER	1
7	978665	HYDRAULIC CYLINDER	1
8	955446	HYDRAULIC CYLINDER	1
9	972115	HYDRAULIC CYLINDER	1
10	LFTY - 100542	BOLT, CYLINDER	4
11	LFTY - 100111	BLOCK, CYL CLAMP	2
12	LFTY - 100119	MOUNTING PLATE	1
13	AGS5588	M24x0.8 X 80.0, SOC HD SCR	8
14	AGS7443	M22x1.5, HEX NUT	16
15	AD24000	M12x1.75 X 30.0, SOC HD SCR	12
16	LFTY - 100677	PAD, INDEXER CLAMP	2
17	LFTY - 100221	BRONZE BUSHING	4
18	AF78665	M24x0.8 X 80.0, SOC HD SCR	8

TUDORTECH DESIGN
Engineering Consultancy Company

HOLD UP ASSEMBLY

LFTY - 100657

A2

STEPS TO CREATE ASSEMBLY DRAWING

1. ***Analyze*** geometry and dimensions of all parts in order to understand the *assembly steps* and overall shape of device or machine.
2. Select an appropriate view.
3. ***Choose major parts***, i.e. parts that have several parts assembled on.
4. Draw a view of ***major parts*** according to a selected viewing direction.

STEPS TO CREATE ASSEMBLY DRAWING

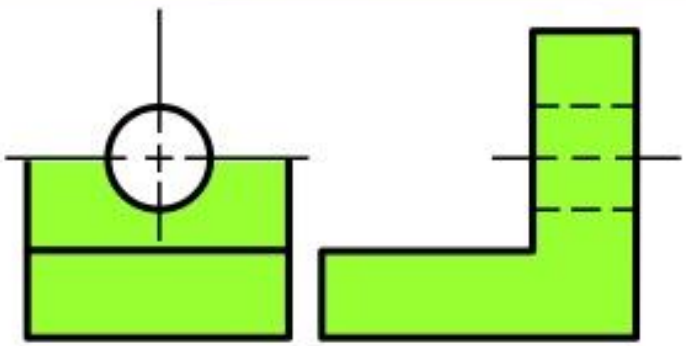
5. Add detail view of the remaining parts at their working positions.
6. Apply ***section technique*** where relative positions between adjacent parts are needed to clarify.
7. Add ***balloons***, ***notes*** and ***dimensions*** (if any).
8. Create BOM.

GENERAL PRACTICE

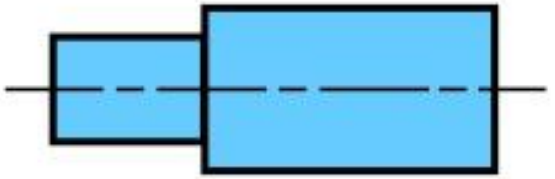
- The ***number of views*** can be one, two, three or more as needed, but it should be **minimum**.
- A good ***viewing direction*** is that represents all (or most) of the parts assembled in their working position.

EXAMPLE : Selection of a necessary view

Given

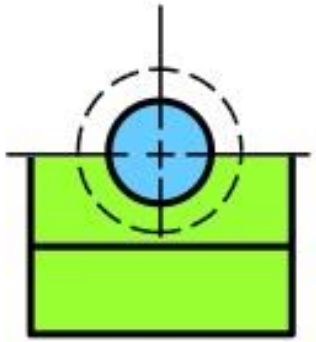


Part A

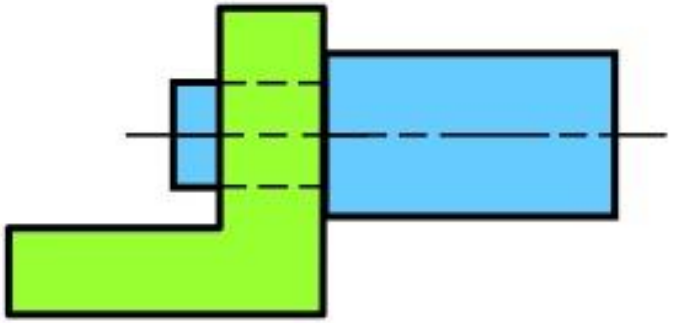


Part B

Student A



Student B



Which is an appropriate view for assembly drawing ?

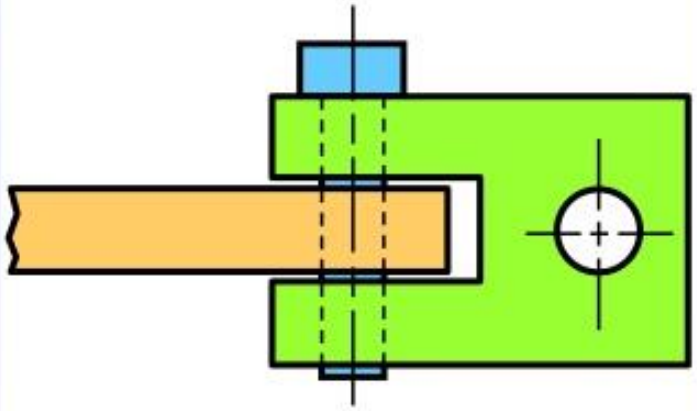
GENERAL PRACTICE

- ***Hidden lines*** usually **omit** unless they are absolutely necessary to illustrate some important feature that the reader might otherwise miss.

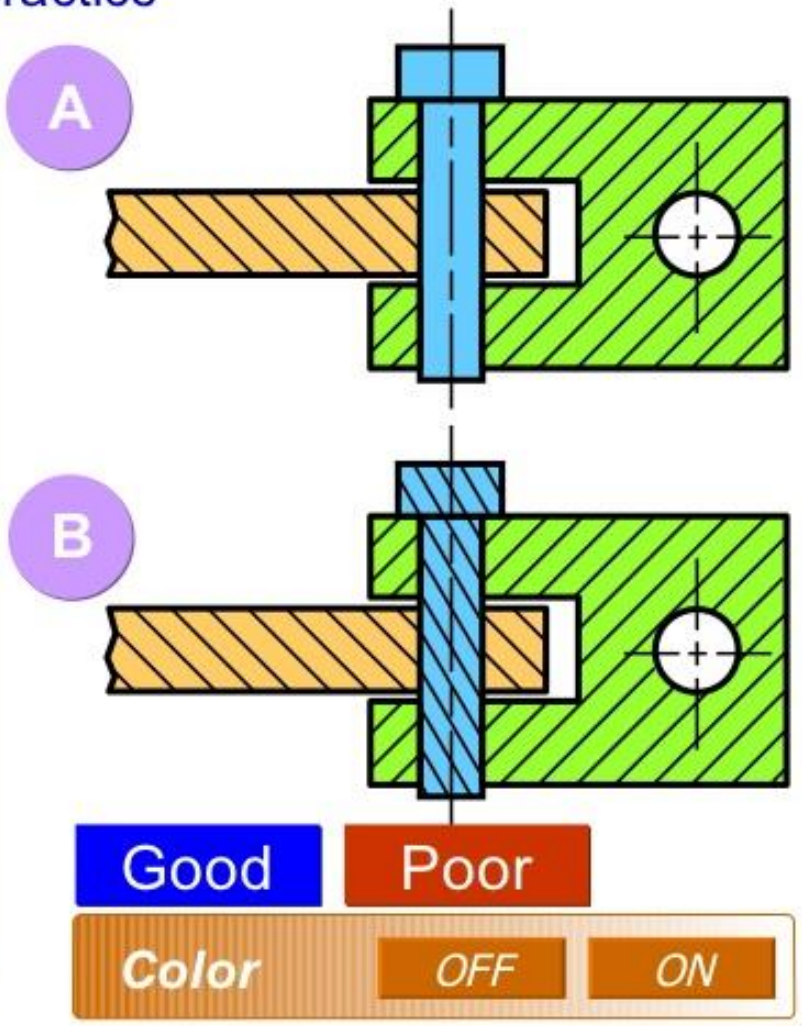
SECTION LINE PRACTICE

- ***Do not*** draw section lines on sectional view of standard parts.
  - Threaded fastener
  - Washer
  - (longitudinal cut of) Solid shaft, Pin, Key

EXAMPLE : Section line practice



Which is an appropriate full section view of this assembly ?



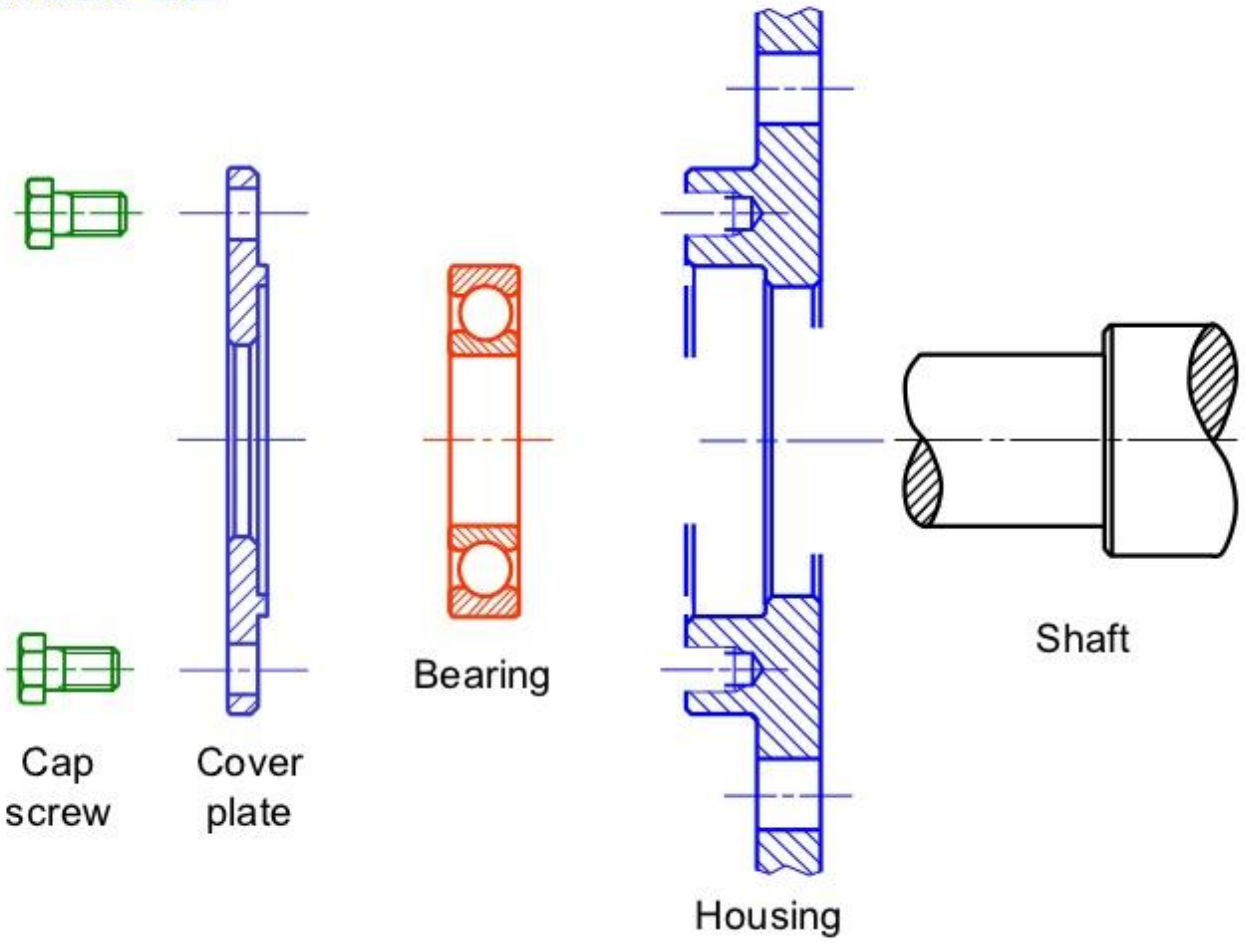
INTERPRETING ASSEMBLY DRAWING

1. Assemble steps.

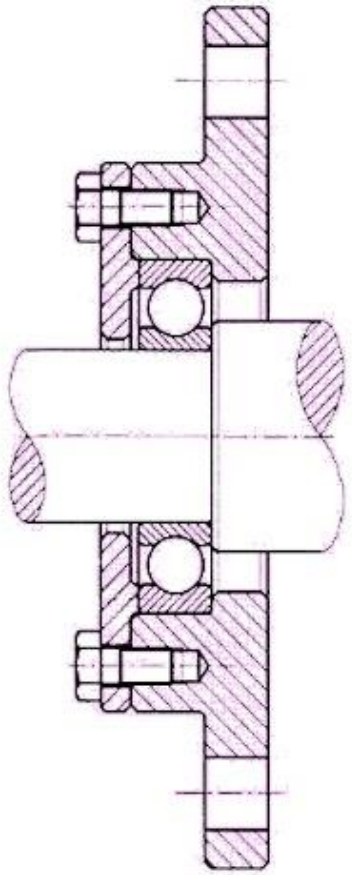
2. Function of each part in machine.

3. Design concept.

EXAMPLE



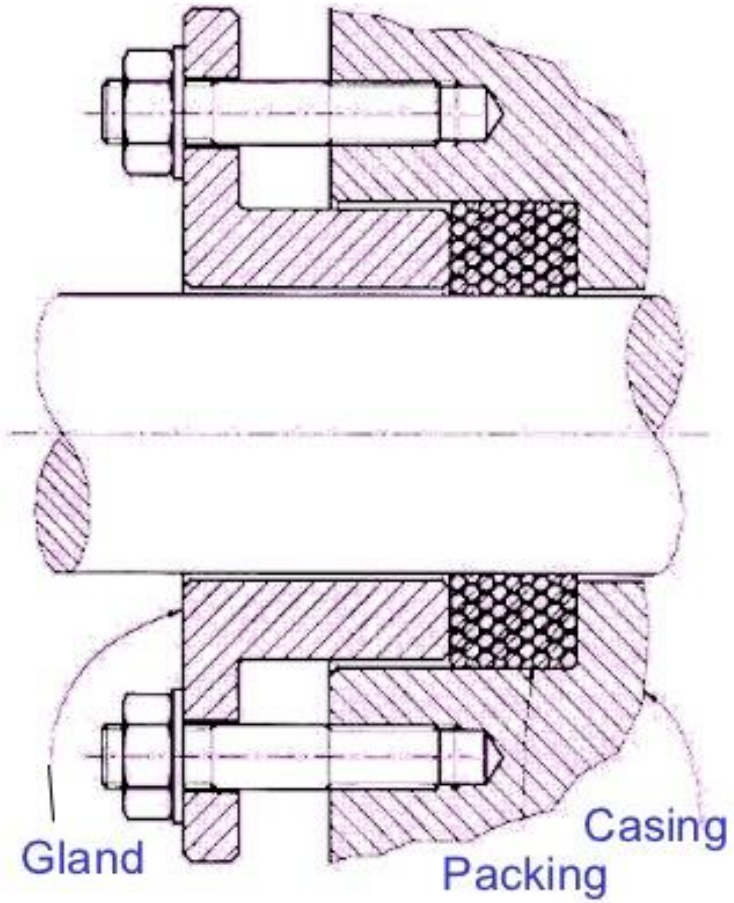
EXAMPLE 1 : *Shaft support on a machine housing*



Assemble steps

1. Install bearing to the shaft.
2. Install the bearing-shaft unit to the housing.
3. Install the cover plate.
4. Tighten the screw.

EXAMPLE 2 : Leakage prevention unit



Function

1. Packing :

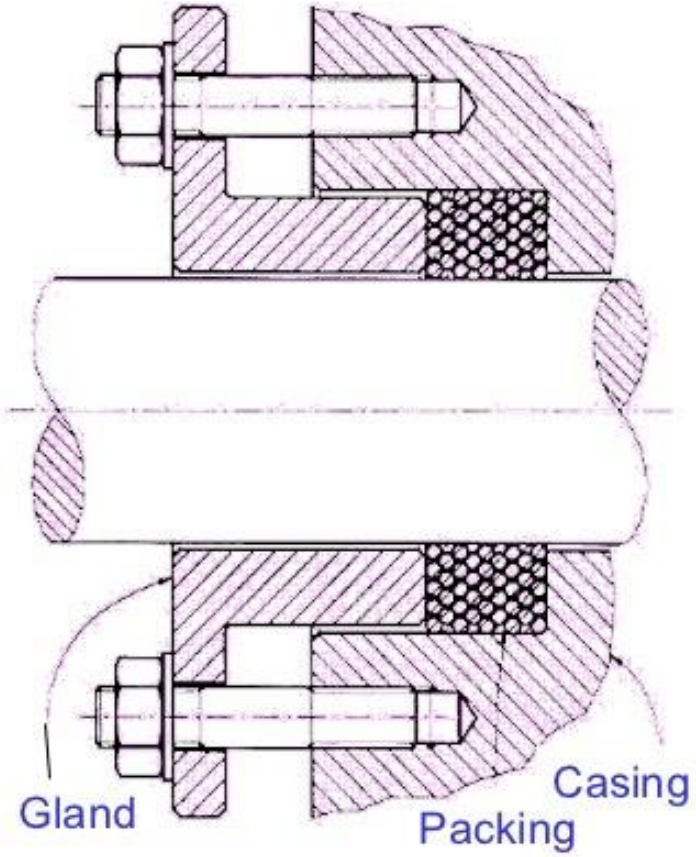
- Preventing the leakage of a fluid inside the casing.

2. Gland :

- Press the packing to make it radial expand and press the shaft surface.



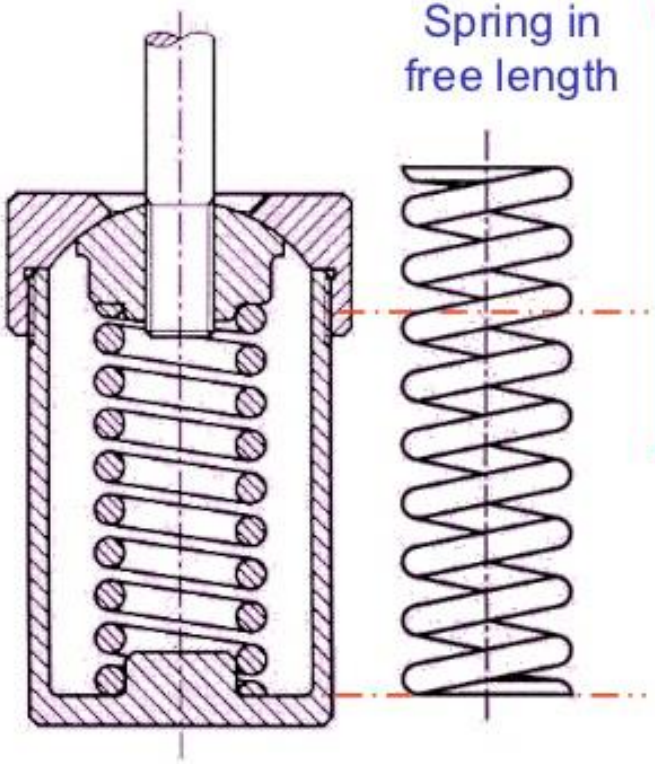
EXAMPLE 2 : Leakage prevention unit



Assemble steps

1. Wrap a packing to the shaft.
2. Install studs to the casing.
3. Install the gland ring where its holes align with stud.
4. Place the washer and tightening the nut.

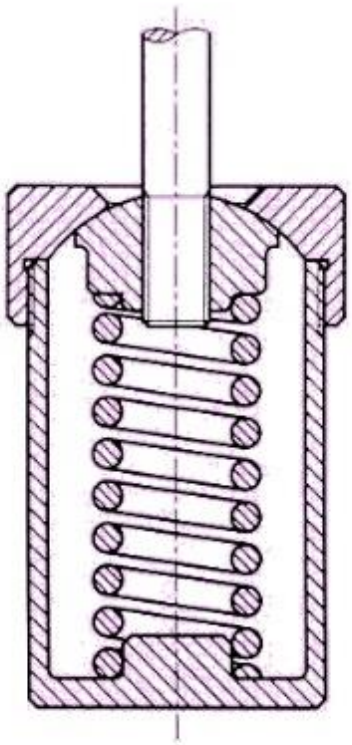
EXAMPLE : *Parts having preloaded spring*



Assemble steps

1. Insert the spring into the casing.
2. Tighten the rod to the spring loader.
3. Close the cap and tighten.

EXAMPLE : *Parts having preloaded spring*

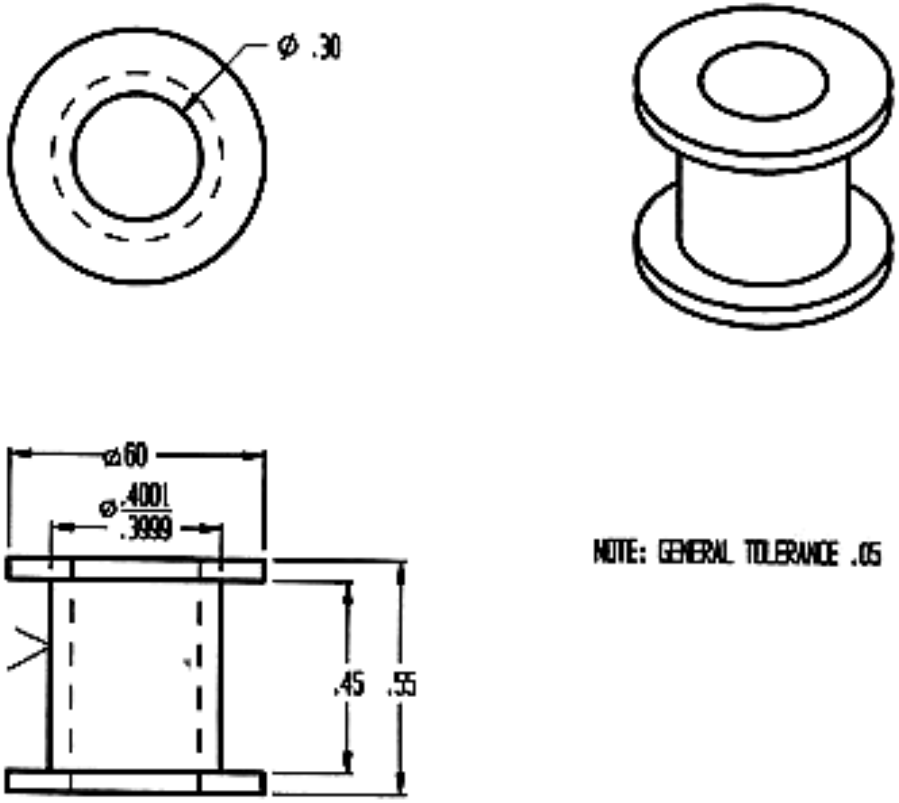


Function

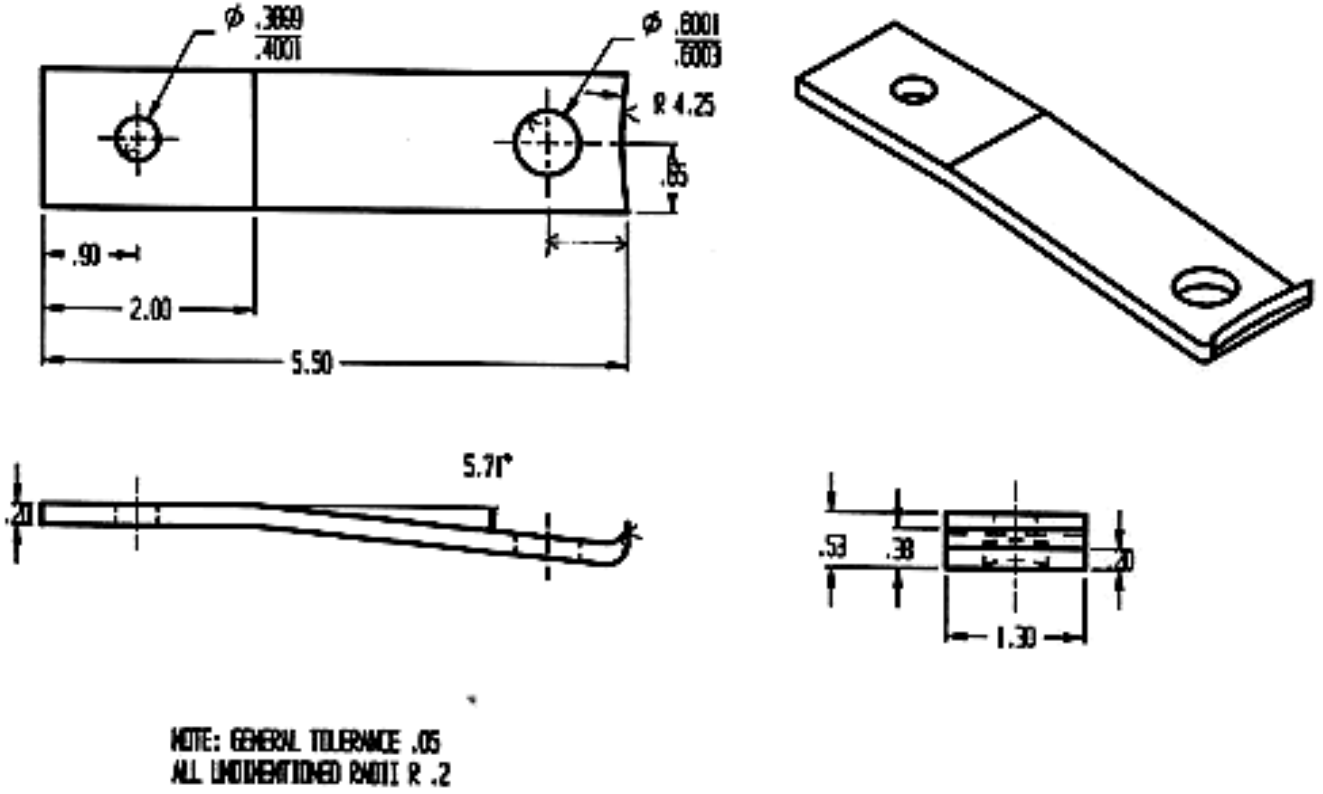
1. Spring plunger :

- Transmit a force from rod to spring.
- Keep the spring in a position.

Detailed Part Drawing: Rivet



Detailed Part Drawing: Bottom Clipper



THE OHIO STATE UNIVERSITY ENGINEERING GRAPHICS DEPARTMENT	DATE: 10-30-94	FILE: CLIPPERF	NAME: PHELPS, BLANCHARD CLEMENS, DELAWARE, OHIO DWG 1 OF 8
TITLE: CLIPPER BOTTOM	SECTION: ME - 10:30 - TEAM 2		